

Graphical Abstract

Multi-Modal Deep Learning in Autonomous Clinical Diagnostics: A Comprehensive Review of Architecture, Fusion Strategies, and Deployment Challenges

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A conceptual schematic illustrating multi-modal diagnostic integration: heterogeneous input channels (CT/MRI radiology images, tabular EHR clinical records, genomic variant profiles, and real-time ICU physiological waveform streams) pass through modality-specific neural encoders into a shared latent space governed by multi-head cross-attention. The unified representation feeds downstream clinical heads for disease classification, survival risk prediction, and automated diagnostic report generation.